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An explainable Ensembled Boosting strategy for making a multi-class intrusion detection framework based on the CICIDS2017 dataset.

Sudip Kumar

Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.

Md. Abdul Based

Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.

Md Rafidul Islam

Dept. Of Computer Science and Technology, Ulster University

Jahanur Biswas

Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.

Md. Nahid Hasan*

Dept. of CSE, Dhaka International University, Dhaka, Bangladesh,

Corresponding author: nahid.hasan.bondhan@gmail.com

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The intrusion detection frameworks play an important role in protecting modern network flow infrastructures to counter progressively mutated cyber-attacks. Many datasets exist, but most are highly imbalanced. In this study, we used the CICIDS2017 dataset to develop a structured and explainable multi-class intrusion detection framework. 20% of the full dataset was used to ensure computational efficiency while maintaining class diversity. In the data preprocessing step, we remove redundant data and handle infinity or missing values, scale features, and correct class imbalance using SMOTE. Here, six machine learning & deep learning models, such as the XGBoost, Logistic Regression (LR), Multi-Layer Perceptron (MLP), Random Forest (RF), K-Nearest Neighbors (KNN), and LightGBM were trained and estimated under the same identical conditions. Performance was predicted under numerous evaluation metrics, including the accuracy, the balanced accuracy, precision, recall, weighted F1-score, MCC, & ROC-AUC. Among six models, the XGBoost model achieved 99.95% accuracy and near perfect ROC-AUC on the test set, the best performance. Then SHAP analysis was applied to enhance transparency. It increases the explainability of the result and makes it trustworthy. Finally, the statistical validation test further ensured the robustness of the result. The proposed framework and the result proved that ensemble boosting with explainable/interpretable artificial intelligence (XAI) provides an accurate and explainable solution for real multiclass intrusion recognition systems.

Keywords: IDS, class imbalance, multi-class classification, cicits2017 dataset, XGBoost, Ensemble Learning, SMOTE, XAI, SHAP, machine learning.

1.1 INTRODUCTION

The speedy development of the internet, modern networks, IoT devices, & cloud computing has made them increasingly vulnerable to cyberattacks. So, intrusion detection systems are widely used to identify malicious behaviors, protect network infrastructure, and monitor traffic. But designing a reliable and perfect intrusion detection system remains challenging due to the large volume of traffic data, the need to detect multiple attack types, and the complexity of threat patterns.

Here, DL and ML models [1] have become more popular than traditional approaches

because they can learn patterns of attack automatically from traffic data and make a preventive system. Some model such as XGBoost, have shown high accuracy in detecting attacks in multi-class intrusion detection [2]. In intrusion environments, DL models have also been successfully adopted to capture complex patterns in attack traffic [3].

Even many papers report high accuracy. But it still has two important challenges. First, IDS datasets are usually extremely imbalanced. Some normal traffic and attack classes have many samples, while others have very few samples. This type of imbalance may cause biases toward the majority class, leading it to perform poorly on minority classes. Here, researchers [1] often use data balancing methods such as SMOTE to overcome these types of problems. Predictions of some high-performing ML and DL models make it difficult to understand how they make decisions. Due to the lack of visibility, the acceptance of cybersecurity applications has decreased. So explainable AI methods like SHAP are used to show how different features invoked model predictions to improve interpretability [2, 3].

To fix these problems, this study proposes a structured and clear multi-class classification intrusion detection system on the CICIDS2017 dataset. We applied SMOTE for managing large data sizes. Through the SMOTE technique, we cleaned, encoded, and balanced the samples and applied them only to the training set, which also avoids data leakage. Another thing is that this study maintains multi-class classification. We use six ML & DL models, among them XGBoost ensured the highest test accuracy, which is 99.95%, and the second highest accuracy 99.91% was achieved by Random Forest (RF). Then the MLP model performed well with 99.70% accuracy, and the KNN model also performed 99.51% accuracy. Lastly, LightGBM and Logistic Regression performed 97.63% and 94.93% accuracy. These results proved that advanced and ensemble models can perfectly detect several types of attacks with very high reliability and precision.

Additionally, SHAP is used for describing the predictions of all models by identifying the most crucial attack traffic patterns. Eventually, statistical tests are conducted to ensure performance differences are truly significant, comparisons are trustworthy and reliable.

Our Key Contributions:

1. Established a multi-class intrusion detection system.
2. Evaluated models using MCC, Cohen's kappa, Balanced accuracy, and G-Mean type comprehensive metrics
3. Integrated SHAP-based interpretable XAI technique to interpret decisions of the model and identify traffic features.
4. Performed the Friedman test to validate whether differences in performance were trustworthy.
5. Provided important visualization: Confusion Matrix, ROC curves, Radar plots & Class distribution analysis.

1.2 RELATED WORK

Due to the rapid growth of the internet and IoT-based systems, cyber-attacks have become more advanced, increasing the demand for more effective intrusion detection frameworks. As attack patterns have become more advanced, traditional security strategies are not suitable in modern times. Now, many researchers want to develop machine learning and deep learning focused IDS to achieve a satisfactory outcome because these algorithms learn attack patterns dynamically. The following section reviews these types of existing ML and DL approaches for intrusion detection.

Mebawondu et al. [1] presented an ensemble deep learning model of a multiclass classification detection system using DBN, CNN, and LSTM-RNN models on the UNSW-NB15 dataset. Their model successfully achieved 0.60 recall, 0.605 precision, and 0.655 F1-score. Their work also addressed class imbalance issues. Hooshmand et al. [2] proposed a smart IDS model that balances samples with select key features by using XGBoost classification techniques with proper SHAP analysis on UNSW-NB15 & NSL-KDD datasets. Their study achieved over 99% and 97% accuracy in binary and multiclass classification. S. Sadhwani et al. [3] worked on making an explainable deep learning focused intrusion detection environment by using LSTM, CNN & Bi-LSTM. They tested on four datasets and got up to 98% accuracy in the CNN model. They used SHAP analysis to make it more trustworthy in real-life scenarios. Gupta et al. [4] work in intrusion detection to improve multiclass classification with managed imbalanced data by using an ensemble deep learning model. The model was tested on CICIDS2017 and NSL-KDD datasets. They used cross-validation for comparative analysis. Adil et al. [5] combine tree-based models (CNN + LSTM + Bi-LSTM + SHAP) and deep learning models with the help of a meta model to propose Ensemble Guard. They used CICIDS2017, UNSW-NB15, NSL-KDD dataset, and got an accuracy of around 98 to 99% with maintain more consistency than the standalone model. Widodo et al. [6] reports SMOTE-based IDS combined with feature selection and log normalization. They used NSL-KDD & CICIDS2017 and proved that SMOTE significantly improves precision, recall & F1-Score with LR, SVM, Naïve Bayes, DT, RF, and XGBoost, achieving an accuracy of up to 99.89%. B. Ncir et al. [7] Build an Explainable ensemble learning IDS that combines three LSTM models. They used SHAP for explainability and tested on UNSW-NB15 and NSL-KDD datasets and achieved above 97% accuracy. The strategy improves both intrusion detection and model transparency performance. Soflaei et al. [8] showed a CTGAN-based strategy for handling the imbalanced data of UNSW-NB15. They trained three shallow classifiers (LR, DT & GNB) followed by two stage label wise ensemble with XGBoost as a meta-classifier and achieved 98% accuracy in binary classification and 95% in multiclass classification. Ahmed et al. [9] Study Report: SHAP-based feature selection IDS that proposed a novel ensemble model method called hybrid bagging and boosting on residuals. The model improves both performance and interpretability and achieved 98.47% accuracy and up to 96% F1-score. Hussein et al. [10] reported in their study that they used the chi-square feature selection method followed by ensemble classifier (AdaBoost, Random Forest & XGBoost) for an intrusion detection framework. They have shown 99.92% accuracy in specific attack, while binary and

multiclass accuracy over 93% and 98%. This study shows ensemble learning model improves detection by reducing feature. Wang et al. [11] reported that their paper proposed an interpretable intrusion detection system using SHAP to provide explanations. They trained two classifiers using fully connected neural networks on the dataset and achieved around 80% accuracy. Narayan et al. [12] showed a hybrid DL based framework using CNN & BiLSTM with proper class imbalance methods on datasets NSL-KDD & UNSW-NB15. They used ROS, ADASYN, SMOTE-Tomek & SMOTE as class imbalancing techniques and got 99.6% & 89.02% accuracy using their two datasets. Ahmed et al. [13] demonstrated an advanced adaptive ensemble model combining three methods (SEM, BMA & CEM). It decreases false positives and adapts to changing attack patterns. This study achieved up to 98.79% on the CICIDS2017 dataset, and SHAP and LIME have been used to achieve proper explainability. Shana et al. [14] proposed machine learning based intrusion detection systems for handling WOA and class imbalance for feature selection. They got the best performance from the Extra Trees model, which was 97% on the UNSW_NB15 and 99.9% on NSL-KDD. Kamil et al. [16] report on their study that's a hybrid intrusion detection framework using RFE_XGBoost and CNN for feature selection, and used SMOTE_BGMM for addressing class imbalance. They used the UNSW-NB15 dataset and achieved 98.80% & 96.49% accuracy in both binary classification and multiclass classification.

After reviewing some existing works, we have clearly seen that some researchers have already used deep learning, machine learning, and ensemble methods to detect network intrusion. But some existing work has a wide evaluation gap among multiple models. Among them, many studies did not address model explainability. These limitations increase the demand for a framework that not only achieves a strong detection accuracy but also addresses fairness in transparency and evaluation while making decisions for intrusion detection. The lack of existing studies motivates the development of the proposed multi-class intrusion detection framework.

1.3 METHODOLOGY

This section of the paper describes complete environmental systems developed for multiclass intrusion detection systems (IDS) using 20% of the CICIDS2017 dataset. The strategy is designed to ensure reproducibility, technical rigor, and transparency. The overall system architecture, from data preparation to explainable models' output, is depicted in Fig. 1.1.

The proposed framework is designed by following a structured pipeline consisting of four major segments:

1. Data Preparation
2. Model development
3. Analysis & Validation
4. Final Output generation.

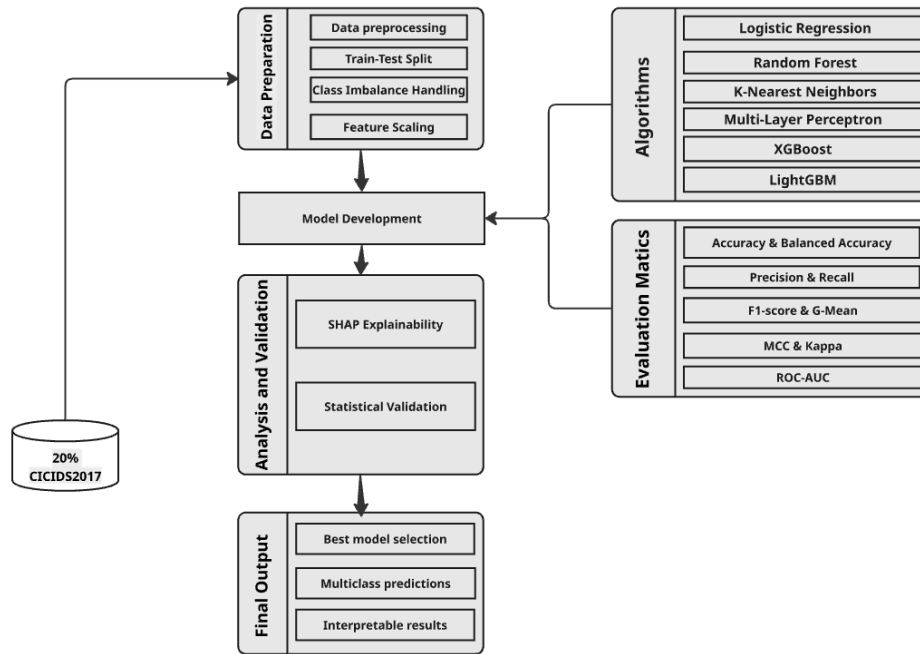


Figure 1.1 Overall workflow of the proposed framework

At first, the system extracted the CICIDS2017 dataset. To ensure computational efficiency and reduce operational time, 20% of the complete CICIDS2017 dataset was used for experiments. This subset included all traffic features, such as DoS, DDoS, normal Traffic, web attacks, brute force, port scanning, and bots.

Many processing steps were applied during the data processing step to improve data quality & enhance model performance. Removed constant and redundant features and handled missing or infinite values appropriately. Encoded categorical labels as numerical values and applied feature standardization to normalize all data. The dataset showed strong class imbalance. SMOTE was used on training data after performing an 80:20 train-split to fix the imbalance problem, prevent information leakage, and ensure fair evaluation.

Then six ML & DL models were implemented in the development stage, such as the Logistic Regression (LR), the Random Forest the K-Nearest Neighbors, the Multi-Layer Perceptron, the XGBoost & the LightGBM. After evaluating model performance, SHapley Additive Explanations (SHAP) was applied for explainability performance. SHAP provides overall feature importance & specific interpretation. After that, statistical validation (like the Friedman test) tests were applied to ensure the performance difference. Finally, the highest-performing model was designated for development in real-time intrusion detection circumstances.

1.3.1 Dataset Description

At the time of selecting datasets, we reviewed many datasets, including old and current datasets, to make an advanced detection system. UNSW-NB15, TII-SSRC-

23, Edge-IIoTset, NF-BoT-IoT, CSE-CIC-IDS2018, and CICIDS2017 were reviewed. After reviewing all datasets, we selected the CICIDS2017 dataset for this research because it is well-structured and provides clear label network traffic and variants of common attack types. Therefore, we used CICIDS2017 for conducting experiments, and the main reason for using 20% of the full dataset is to ensure computational efficiency and reduce operational time. The dataset was collected from Kaggle [15].

1.3.2 Data Preprocessing

Many preprocessing methods were used to ensure reliable and stable model training. Initially, redundant and constant value features were removed to handle unnecessary information. we carefully handled infinite and missing values during training. Then, encoded attack categorical levels in a numerical form to enable multiclass classification. Finally, Feature Scaling was applied through Standard Normalization to ensure all numerical columns were on a comparable scale.

1.3.3 Class Imbalance Handling

The dataset showed notable class imbalance, mainly for minority attack categories. The imbalance is illustrated in **Figure 1.2** before correction. To fix this problem, SMOTE strategy was used in the training data only, as suggested by Kamil et al. [16]. After applying SMOTE, the distribution class became balanced, as depicted in Fig. 1.2.

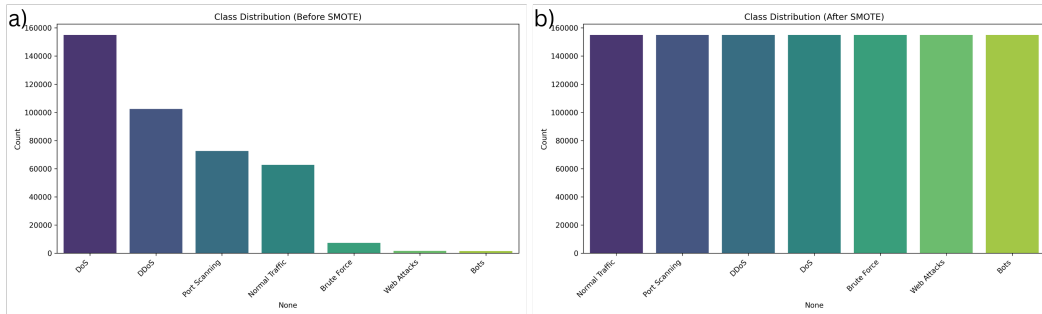


Figure 1.2 Class distribution before and after SMOTE

1.3.4 Model Development

Six models were implemented in this study. Models were trained through the same identical preprocessing steps, including 80:20 train-test split, balancing & feature standardization. To get a fair and unbiased performance, we used the same preprocessing pipeline for all classifiers.

1.3.4.1 Logistic Regression

Logistic Regression was applied as a baseline classifier, as suggested by Widodo et al. [6]. Using the Sigmoid Function in Equation 1.1, it guesses the chance of class membership:

$$p(y = 1 | x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (1.1)$$

Where w and b represent the weight vector and bias term. The SoftMax function was applied for multi-class classification so that the model can be extended to multiple categories.

1.3.4.2 Random Forest

Random Forest is used as suggested by Widodo et al. [6] for multiple decision trees, which use bootstrap sampling and combine their prediction results through majority voting. Gini Impurity measure is used by the RF algorithm in equation 1.2 during the tree construction process.

$$G = 1 - \sum_{i=1}^n p_i^2 \quad (1.2)$$

Where p_i denotes the probability of class i at a given node. Random forest decreases variance, mitigates overfitting, and improves generalization by aggregating multiple trees.

1.3.4.3 K-Nearest Neighbors

KNN is a distance-based ML algorithm that is used for assigning a class label [?]. It was used with the Euclidean distance metric to measure similarity between distances using Function 1.3:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2} \quad (1.3)$$

Where n represents the number of features. K-Nearest Neighbors captures local data structure and is sensitive to feature scaling, making standardization essential.

1.3.4.4 Multi-Layer Perceptron

MLP is a neural network (NN) that consists of a fully connected hidden layer [?]. It works as ReLU Function 1.4:

$$f(y) = \max(0, y) \quad (1.4)$$

This ReLU activation function doesn't change positive input values and outputs

zero for negative input values. It helps in the complex pattern of network learn and improves computation efficiency.

1.3.4.5 *LightGBM*

The LightBGM machine learning model is implemented as suggested in an existing study [17] that represent a powerful gradient boosting model. It makes decisions on trees serially, where new trees are trained by the previous trees to detect errors. To decrease computational losses, it uses leaf-based tree growth approaches, and to speed up and reduce memory waste, it also applies histogram-based learning.

1.3.4.6 *XGBoost*

The XGBoost model of machine learning is also implemented by a recommendation of a previous existing study [2] that represent an advanced gradient boosting algorithm. It also works like LightGBM model to reduce classification errors by building trees sequentially. However, to balance complexity and model accuracy and prevent over-fitting, this model includes a regularized objective function. This mechanism helps to improve generalization performance on new unseen data.

1.3.5 Model Evaluation Strategy

Since the dataset contains imbalanced classes, some additional metrics were implemented to evaluate performance properly for both minority and majority classes. We have used many different metrics instead of relying on one metric to ensure fair assessment. It evaluated both common attack types and minority attack categories properly. Using multiple metrics gives a fair understanding of how all models performed well for all classes.

1.4 RESULTS & DISCUSSION

This part exhibits experimental results from the comparative evaluation of all algorithms based on the CICIDS2017 dataset. These results integrate performance metrics, interpretability insights & statistical validation . The rateables comparison of six models on the test dataset is evaluated in Table 1.1.

Table 1.1 Test set performance

Models	Accuracy	Balanced Acc.	F1-Score (W)	MCC	ROC-AUC (W)
LR	0.9494	0.9563	0.9574	0.9328	0.9972
RF	0.9991	0.9954	0.9991	0.9988	0.9999
KNN	0.9951	0.9909	0.9951	0.9933	0.9983
MLP	0.9971	0.9956	0.9971	0.9960	0.9999
LightGBM	0.9763	0.9400	0.9790	0.9678	0.9798
XGBoost	0.9995	0.9989	0.9995	0.9993	0.99998

According to Table 1.1, **XGBoost** achieved the highest performance. This

achieved 99.95% accuracy, 99.88% balanced accuracy, a weighted F1-score of 99.95%, a Matthews correlation coefficient of 0.9993, & an ROC-AUC of 0.99998. That denotes near-perfect classification performance among both majority classes and minority classes.

MLP and Random Forest (RF) also performed strongly, achieving accuracy values over 99% mentioned in Table 1.1. KNN performed comparatively, but a little below the boosting-below model. Logistic regression (LR) and LightBGM showed comparatively lower accuracy values, which indicates decreased performance on minority attack classes. To illustrate the metric-wise comparison, a bar plot for accuracy, a balanced accuracy & weighted F1-Score is illustrated in Fig 1.3:

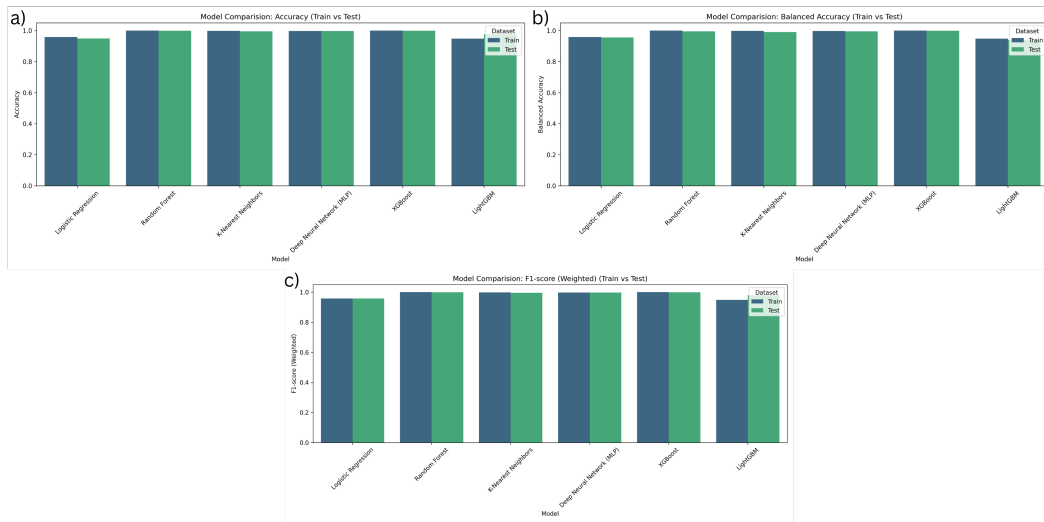


Figure 1.3 Comparison of accuracy, balanced accuracy, and F1-Score

These figures confirmed the superiority boosting based ensemble models, particularly the XGBoost, among multiple performance dimensions.

1.4.1 Confusion Matrix Analysis

The confusion matrices show a strong diagonal dominance for e) XGBoost & b) Random Forest (RF), which indicates a high classification among all attack categories in Figure 1.4 Finally, a) Logistic Regression (LR) model demonstrated slightly higher confusion between near-related attack categories, which explains its lower balance accuracy. f) LightBGM model also shows minor inconsistencies in minority class classification.

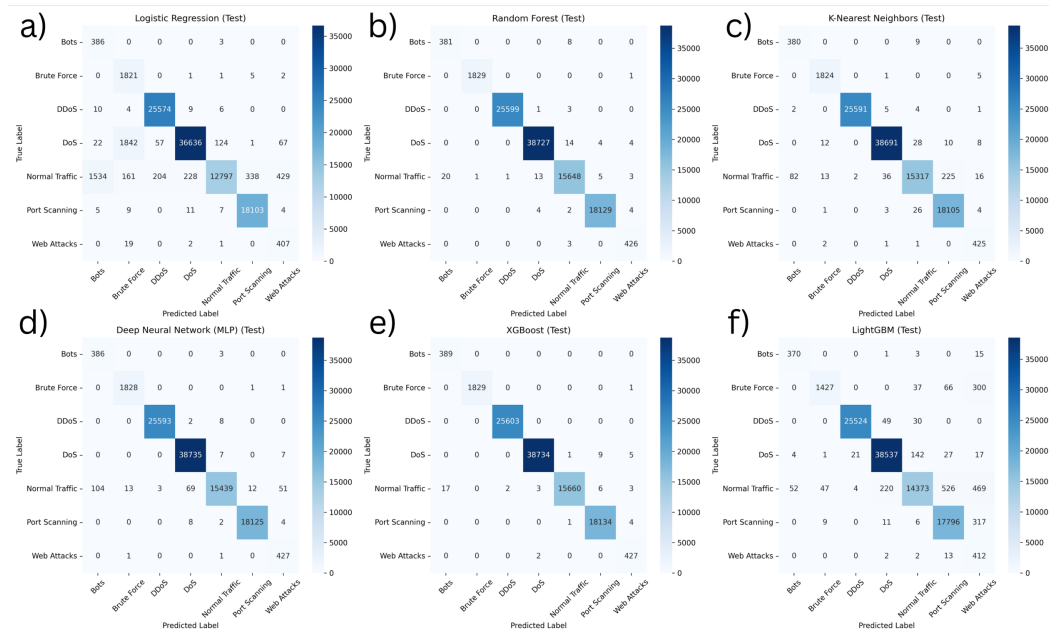


Figure 1.4 Confusion matrices of all models

1.4.2 Curve Analysis

ROC curves were generated to evaluate class separability & generalization performance:

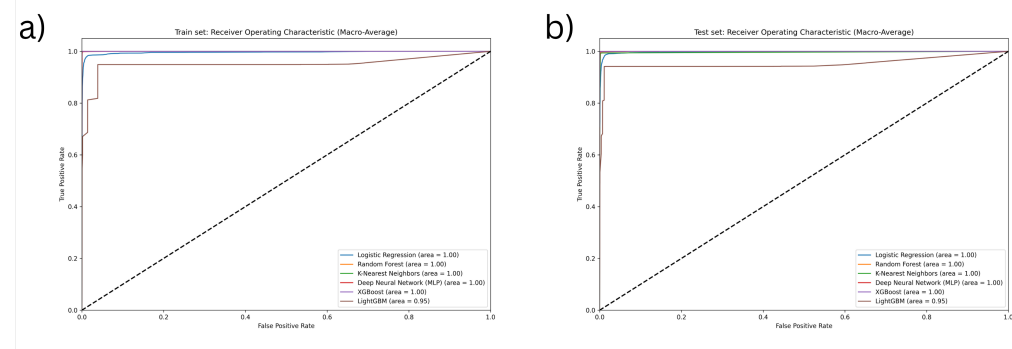


Figure 1.5 ROC curve of the Train and Test sets

The curves indicate that the XGBoost model achieved close to perfect values on the test set, around 0.99998, which confirmed its super ability to distinguish between different traffic classes. MLP and Random Forest also achieved high AUC values, while LightBGM and Logistic Regression showed slightly lower separation AUC values.

1.4.3 Metric Stability and Robustness

Radar plots were created for each model to assess robustness according to evaluation metrics depicted in Fig 1.6.

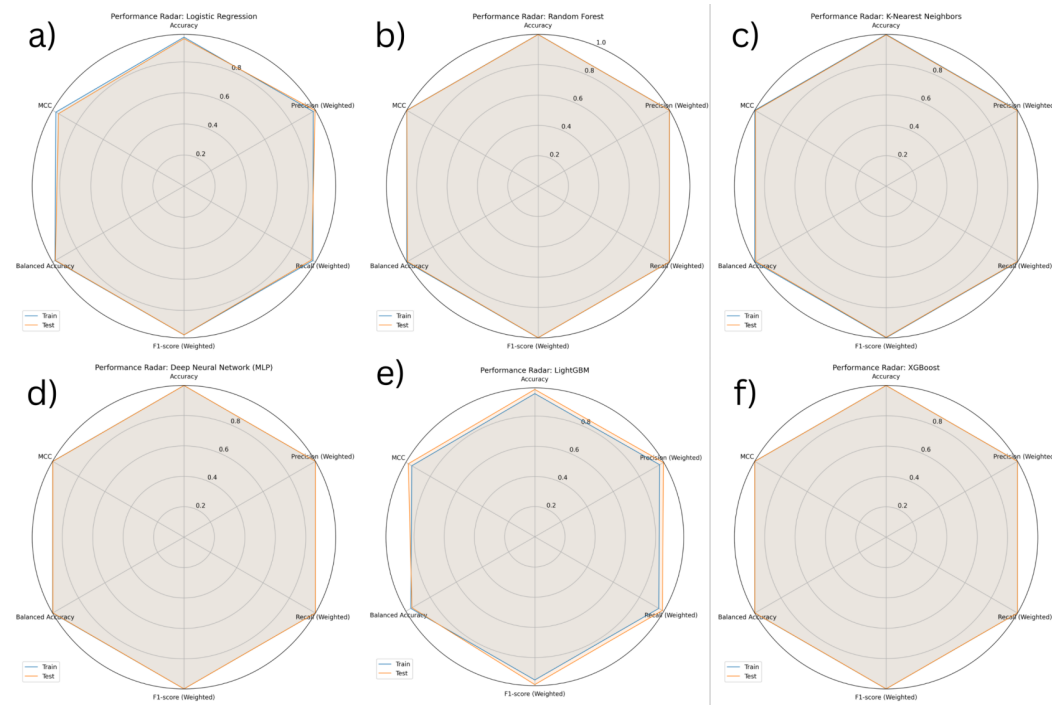


Figure 1.6 Radar plots of all models

All radar plots of matrices indicate that the performance of the f) XGBoost model is better than the other six models across performance metrics, such as accuracy, F1-score, and so on. Its performance remained consistent and stable, including reliable and strong classification capability. At the same time, a) Logistic Regression and e) LightGBM have slightly decreased their performance in detecting minority attack classes. It highlights that the gradient boosting method is more capable than others in handling complex relations between varied cyber-attacks and different network features.

1.4.4 Interpretability Analysis Using SHAP

SHAP analysis was applied to enhance the best-performing model transparency, which is XGBoost. The global feature ranking was represented in Fig 1.7a, where features like Destination Port, Init_Win_bytes_forward, Flow Bytes, Flow duration, and Flow IAT Min were calculated to have a strong average impact on prediction. Distribution and directional impact of the features were depicted in Fig. 1.7b, which shows how low and high feature values work negatively and positively to detect attack classification. Feature interaction behavior was also illustrated in the dependence

plot, Fig. 1.7c, which demonstrated that destination port worked with other variables and created an impact to models output flowing nonlinear manner.

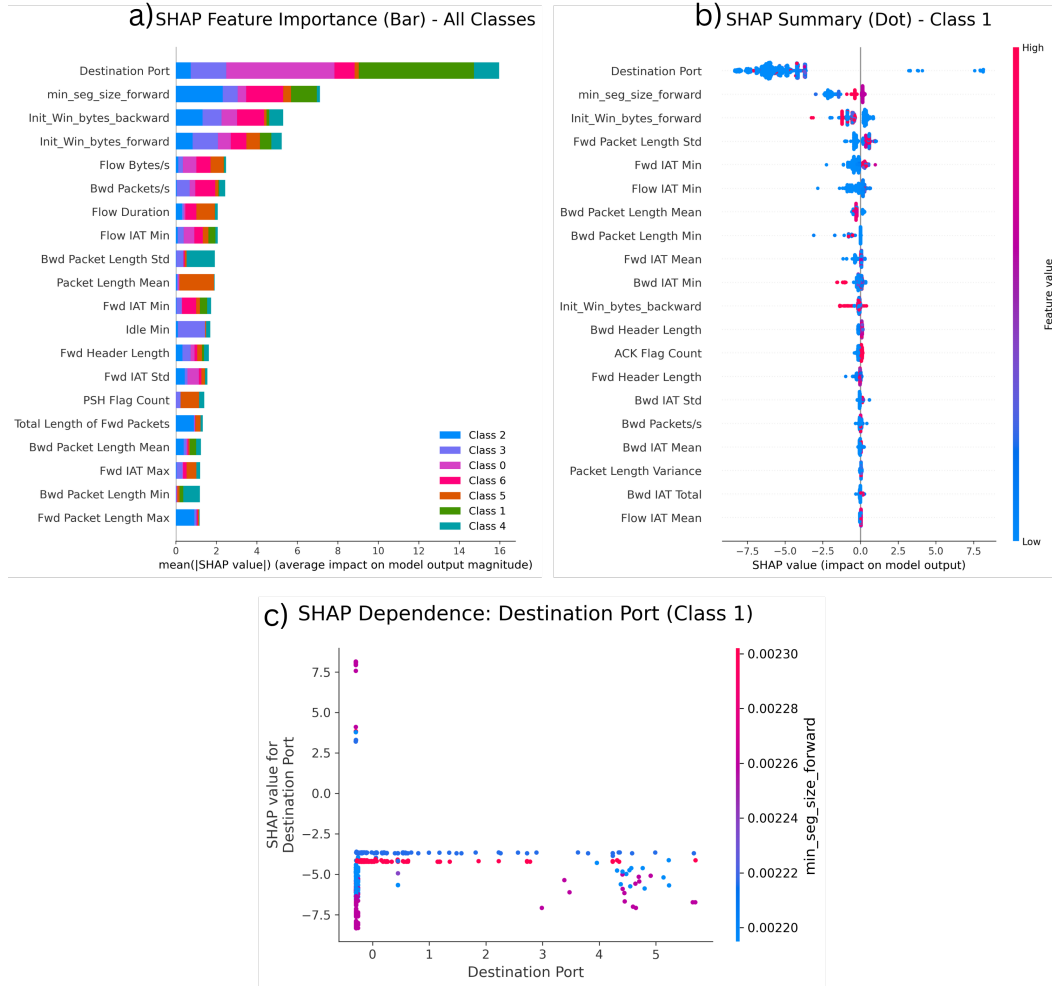


Figure 1.7 SHAP Analysis

The SHAP analysis ensures that the model's predictions were primarily measured by focusing on meaningful network traffic features instead of random correlations. This enhances trust in the model's performance and builds confidence in the model's acceptability, which supports to make suitability for real intrusion detection frameworks.

1.4.5 Statistical Validation

The Friedman statistical test was conducted to identify performance differences that were not random. Obtained a 0.41588 P value that the differences were not statistically significant at the 0.05 level.

If it was smaller than 0.05p, one model had clearly winner. Since the result means p-value is 0.41588, which is greater than 0.05, that means no model is clearly a winner over the other models. All three models, such as XGBoost, MLP, and RF among the six models, did equally well. Therefore, the result confirmed that ensemble models such as XGBoost, RF, and MLP performed under the same experimental conditions, confirming the robustness of the evaluation framework.

1.4.6 Discussion

The results proved that the proposed system performs better than many recent detection methods we have reported in the table below. The table's highest accuracy showed lower than our proposed model's accuracy, which is 99.95% achieved by XGBoost model. This improvement became possible due to the proper preprocessing of data, handling class imbalance using SMOTE, and a valid comparison of six models under the same conditions.

In addition, we used SHAP-based interpretability to prove how the model makes high-accuracy decisions, unlike several previous studies was only focused on prediction accuracy. Therefore, the result suggested that the proposed intrusion detection framework is accurate and reliable for real multiclass intrusion detection systems.

Table 1.2 Comparison with existing works

Authors	Algorithms	Acc.
Mebawondu et al. [1]	CNN + LSTM Ensemble	98.7%
Hooshmand et al. [2]	RF + XGBoost + SHAP	99.2%
S. Sadhwani et al. [3]	CNN + SHAP	97.8%
Gupta et al. [4]	Ensemble Deep Learning model	92.37%, 86.62%
Adil et al. [5]	An explainable meta model that combines tree-based models	99.89%
Widodo et al. [6]	LR + SVM + Naïve Bayes + DT + RF + XGBoost	99.89%
B. Ncir et al. [7]	Ensemble of AdaBoost, RF, XGBoost)	97.05%,95.15%
Soflaei et al. [8]	CTGAN + DT + LR + GNB+ XGBoost	98%
Ahmed et al. [9]	Hybrid Bagging & Boosting + SHAP	98.47%
Hussein et al. [10]	Ensemble (AdaBoost + RF + XGBoost)	99.92%
Wang et al. [11]	Neural Network + SHAP	80%
Narayan et al. [12]	CNN + Bi-LSTM	99.6%,89.02%
Ahmed et al. [13]	SEM + BMA + CEM + SHAP + LIME	97-98%
Shana et al. [14]	SMOTE + Bagging+XGBoost + Adaboost + Extra Trees	99.90%,97%
Kamil et al. [16]	SMOTE + BGMM + CNN + RFE + BGMM	98.80%,96.49%
Proposed Work	XGBoost	99.95%

1.5 CONCLUSION

In this study, a well-structured and explainable multiclass intrusion detection framework was developed using the CICIDS2017 dataset. The proposed strategy combined

data preprocessing, class imbalance handling, and evaluation of six ML and DL classifiers under the same experimental conditions. Overall models, XGBoost achieved the highest balance accuracy, weighted F1-Score, MCC & ROC-AUC values. The result demonstrates that the ensemble boosting model outperforms traditional distance-based and linear models for multiclass classification detection clearly. Beyond performance, this work also ensures transparency by applying SHAP-based interpretability. In addition, the statistical validation test confirmed the reliability and robustness of the comparative evaluation.

Overall, the proposed architecture provides a balanced combination of high accuracy, interpretability, and robustness. These attributes make it reliable for detecting real-world intrusions with high performance and trust. Future work may be scalable for large datasets, integration of advanced DL-based models to improve detection capabilities & real-time streaming implementation.

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